

Runqiu Wang

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EDUCATION

2026 - Ph.D. Student in Computer Science at **University of Wisconsin–Madison**
2024 - 2026 B.S.E in Computer Science at **University of Michigan** (GPA: 3.89/4.0)
2022 - 2026 B.S. in Mechanical Engineering at **Shanghai Jiao Tong University** (GPA: 3.57/4.0)

RESEARCH EXPERIENCE

Undergraduate Research Assistant, AMI Lab Jan. 2025 – March 2026

University of Michigan · Ann Arbor, MI · Advisor: Ke Sun

- Project 1: Practical Moiré-Based Visualization of Airflows with Commodity Cameras
 - Co-first author; designed and implemented a sensing pipeline, using Moiré-patterned wallpaper and low-cost cameras to reveal otherwise invisible heat plumes and gas leaks for real-world applications.
 - Developed a calibration module that aligns the camera to the wallpaper and automatically tunes the pattern frequency to keep a stable Moiré pattern under changing camera poses.
 - Built downstream processing and ML pipeline, collecting 8.25h of data and training PyTorch models for dry/moist-heat classification (99.8% acc.) and oil-temperature regression (MAE $\approx 7^{\circ}\text{C}$).
- Project 2: Moiré-based Ultra-Sensitive Visuo-Tactile Sensor
 - Engineered an automated data collection pipeline to curate a robust dataset of $> 20k$ samples, and drafted evaluation sections and analytical diagrams for the manuscript.
 - Designed and conducted experiments to benchmark the sensor’s perception and force estimation.
- Project 3: Wearable IMU-based Activity Recognition and Motion Transcription
 - Surveyed ~ 50 papers on wearable IMU sensing and reproduced key baselines.
 - Implemented a transformer-based IMU localization prototype for flexible device combinations.

Undergraduate Research Assistant, DIDIS Lab April 2024 – Aug. 2024

Shanghai Jiao Tong University · Shanghai, China · Advisor: Youyi Bi

Project: Transformer-Based Motion Planning via Seq2Seq Generation

- Implemented a loss module to combine configuration, point-cloud matching, and collision losses.
- Built attention and point-cloud visualization scripts to analyze how the model attends to obstacles.
- Ran ablation experiments on learning-rate schedulers to study their effects on path quality.

PUBLICATIONS

Zhu*, Linzhen, **Runqiu Wang***, Yi Rong, and Ke Sun (May 2026). “MoiréLens: Bringing Schlieren Imaging into Real-World Environments using Moiré Patterns”. In: *Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys '26)*. *Equal contribution., pp. 1274–1287. DOI: [10.1145/3774906.3802765](https://doi.org/10.1145/3774906.3802765).

Zhu, Linzhen, Liuxi Xing, **Runqiu Wang**, and Ke Sun (Oct. 2026). “MoiréSkin: Ultra-Sensitive Sensor-Actuator Visuo-Tactile Skin using Moiré Patterns”. In: *Proceedings of the 32nd Annual International Conference on Mobile Computing and Networking (MobiCom '26)*.

Boyan Li, **Runqiu Wang**, Yulin Chen, Bohan Feng, Qi Zhou, and Youyi Bi (Aug. 2025). “Motion Planning Transformers (MP-Former): Treat Motion Planning As Sequence-To-Sequence Generation”. In: *Proceedings of the ASME 2025 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE 2025)*. Anaheim, California, USA. DOI: [10.1115/DETC2025-169215](https://doi.org/10.1115/DETC2025-169215).

SELECTED PROJECTS

Game Dev: Bargain Brawl (Two-Player Party Game) [[Trailer](#)] Jan. 2026 – April 2026

- Programmed core player mechanics, controller inputs, and interactive item abilities (e.g., speed boosts, environmental hazards) for a fast-paced, split-screen multiplayer experience.
- Implemented the end-to-end game flow, UI navigation, and an occlusion-handling camera system.

Network Simulation (*EECS 489: Computer Networks*) Jan. 2025 – April 2025

- Simulated network topologies in Mininet and wrote C++ socket programs to measure RTT and throughput, reproducing bufferbloat and analyzing its impact on latency-sensitive flows.
- Built a video proxy with load balancing and adaptive DASH streaming to improve quality of experience under varying bandwidth conditions.
- Implemented a POX-based SDN controller and a TCP-like reliable transport over UDP, along with an L3 router supporting ARP and ICMP.

AI-Driven AR Drawing Assistant (*EECS 498: AI-Enabled Mixed Reality*) Jan. 2025 – April 2025

- Designed and implemented an AR system that turns voice descriptions into AI-generated line drawings projected onto paper, allowing users to trace custom sketches.
- Implemented a GPT-4o + DALL·E 3 image pipeline and a Meta Quest 3 AR module (Unity + OpenCV/ArUco + solvePnP) for marker-based projection alignment.

Bluetooth-Controlled Robotics Vehicle (*First Place, 13th SJTU "Liming Cup" Mechanical Innovation Competition For Freshman*) March 2023 – May 2023

- Developed an Arduino-based Bluetooth control system using the HC-06 module for PS2 controller.
- Designed circuits integrated with the mechanical chassis; collaborated on assembly and testing.

Robotic Arm with Flexible Grippers (*VM 250: Design and Manufacture*) March 2024 – May 2024

- Led a four-person team in designing and manufacturing a robotic arm with soft silicone grippers.
- Designed and assembled a 5 DOF robotic arm and its circuits to achieve flexible gripping capabilities.
- Developed an Arduino-based control system that connects to a mobile app, enabling remote control.

HONORS AND AWARDS

James B. Angell Scholar, University of Michigan	March 2026
Engineering Continuing Student Scholarship (merit-based), University of Michigan	2025–2026
First Prize, 13th SJTU "Liming Cup" Mechanical Innovation Competition For Freshman	May 2023
UM-SJTU Joint Institute Outstanding Freshman Scholarship	2022–2023

SKILLS

Programming	C/C++, Python, MATLAB, Jupyter Notebook, SQLite, Bash, Unity (C#)
Hardware	Solidworks, 3D printing, Arduino, Raspberry Pi, Mechanical Design

ACTIVITIES

Michigan Hackers, Member	Sep. 2024 – Dec. 2024
Student Union Tech Department of UM-SJTU Joint Institute, Member	Oct. 2022 – Aug. 2023